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Cardiac Considerations in Pediatrics Part I

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Do.
Make.
Innovate.
Reinvent the Future.

Session Objectives

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1. Students will be able to identify the fetal circulation shunts and demonstrate an understanding of the maternal-fetal circulation
2. Students will be able to understand the anatomic, and physiologic basis of the cyanotic congenital heart defects
    - Such as persistent truncus arteriosus, transposition of great vessels, tricuspid atresia, tetralogy of Fallot and total anomalous pulmonary venous return (TAPVR), hypoplastic left heart syndrome , and ebstein's anomaly
3. Students will apply the clinical signs, symptoms, lab and imaging findings and correlate them to cyanotic congenital heart defects
4. Students will be able to apply information from this lecture to board style questions

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Session Outline

1. Maternal-fetal Circulation
2. Cyanotic heart defects
    1. Persistent Truncus Arteriosus
    2. Transposition of Great Vessels
    3. Tricuspid Atresia
    4. Tetralogy of Fallot
    5. Total Anomalous Pulmonary Venous Return
3. A few other defects
    1. Hypoplastic left heart
    2. Ebstein anomaly

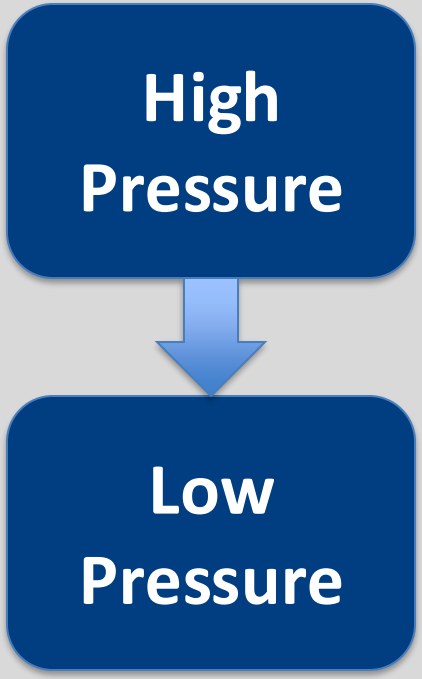
4. Questions



Review: Blood Flow Physics

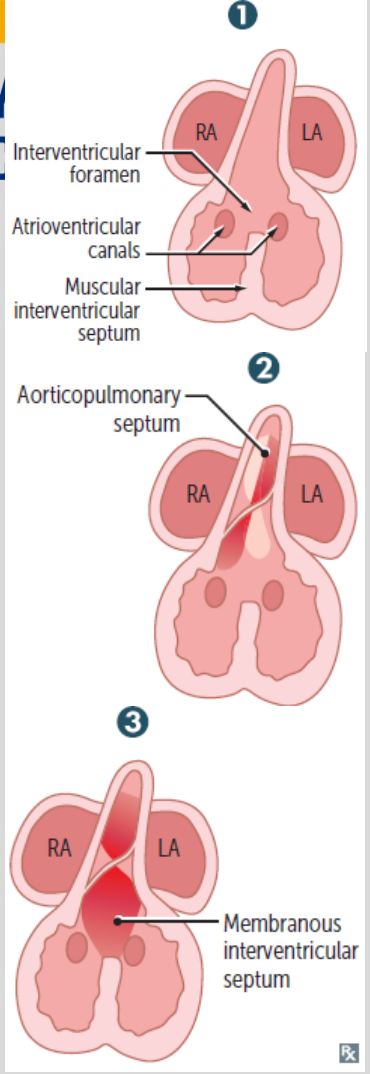
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- Blood flows from high → low pressure
  - Down the pressure gradient
- Areas of higher pressure have more resistance
- Blood takes the "path of least resistance"
- As a fetus, RA pressure > LA pressure
- After birth reverses and Left sided pressures are higher

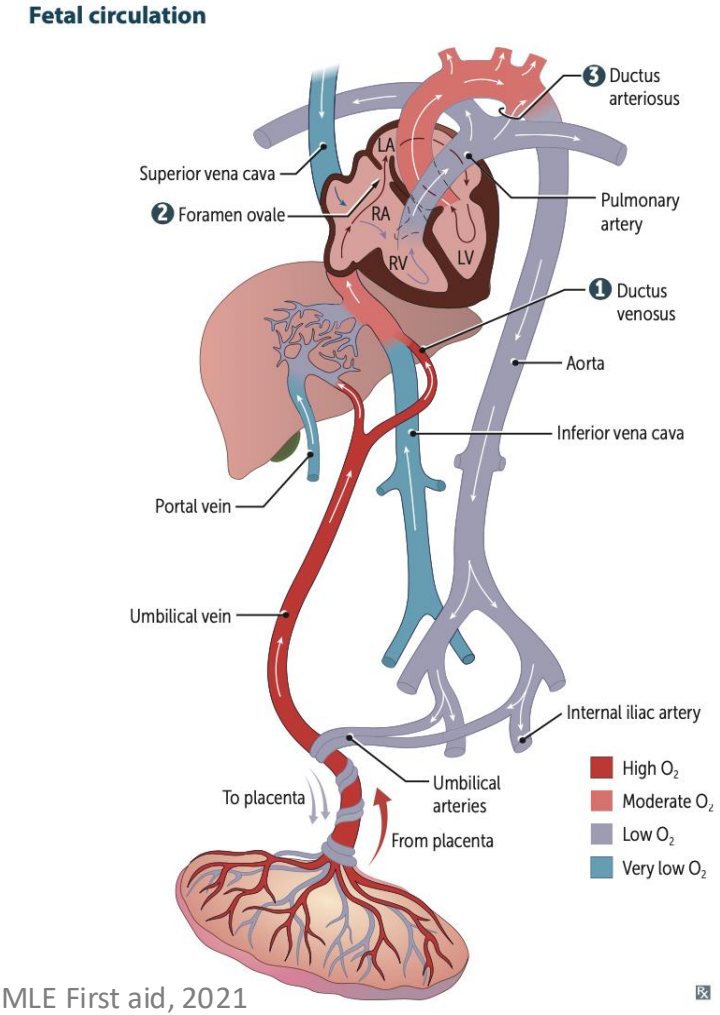


Embryology Review

- **Neural crest and endocardial cells** migrate to form truncal and bulbar ridges → spiral and fuse → form aorticopulmonary septum → ascending aorta & pulmonary trunk
- Failure of neural crest cells to migrate → outflow tract abnormalities



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USMLE First aid, 2021

MOM

Umbilical vein

Umbilical arteries

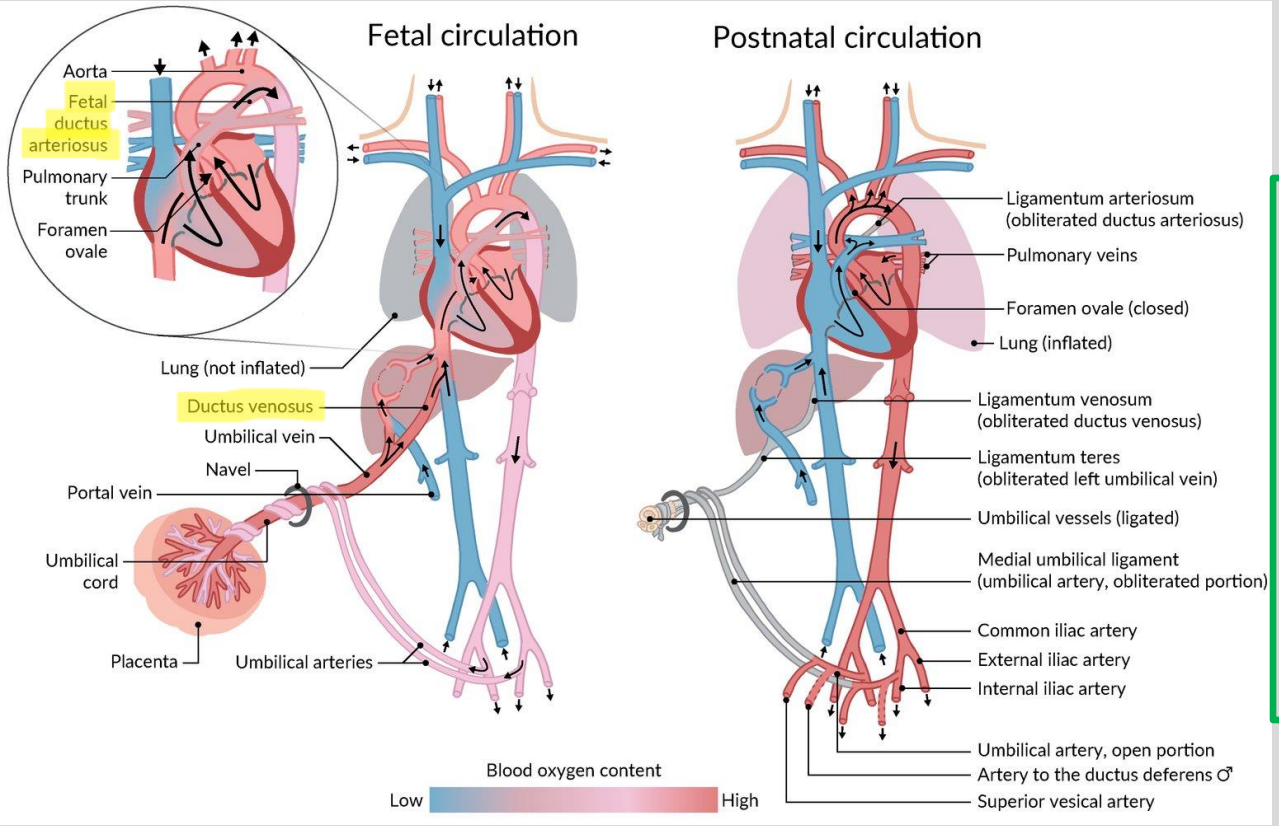
BABY

Umbilical vein brings oxygen from mom to fetus

Umbilical vein = 1

Umbilical ar-TWO-ries

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Ductus Venosus: bypasses liver (umbilical vein → IVC)

Foramen Ovale: bypasses lungs (RA → LA)

Ductus Arteriosus: bypasses lungs (pulmonary artery → aorta)

https://www.amboss.com/us/knowledge/Prenatal_and_postnatal_physiology/

Changes at Birth

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- Prior to birth, lungs not inflated and have high pressures
- First breath →
  - Lungs expand & ↓ pulmonary vascular resistance & pressure → ↑ pulmonary blood flow → ↑ LA pressure



Shunt Closures

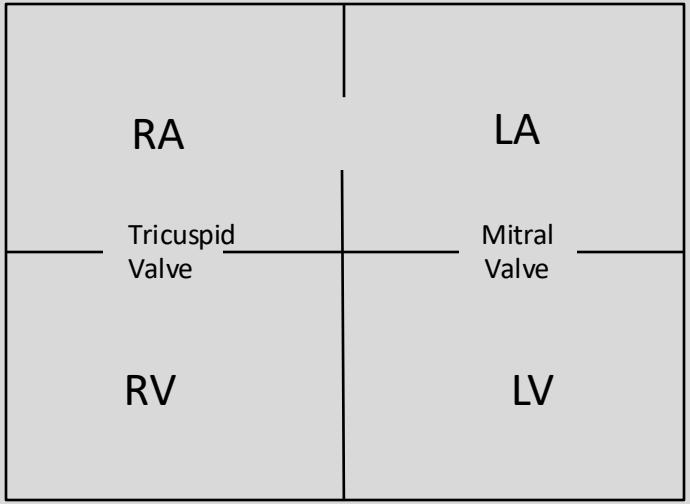
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Shunt → Remnant	Cause of Closure	Result of Closure
Ductus venosus → Ligamentum venosum	Umbilical cord is clamped → absent blood flow through umbilical vein → closure	
Foramen ovale → Fossa ovalis	LA pressure > RA pressure → LA pressure causes septa to press against each other	↑ blood to pulmonary trunk because pulmonary vascular resistance < systemic resistance
Ductus arteriosus → Ligamentum arteriosum (1-5 days)	↑ oxygen & ↓ prostaglandins → closure of Ductus arteriosus	↑ blood flow to lungs

Overview of Cyanotic Congenital Heart Defects

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Right to Left Shunts



R → L shunt = blue babies ea**RL**y

L → R shunt = problems **L**ate**R**

The 5 T's

1. **T**runcus arteriosus (1 Vessel)
2. **T**ransposition of Great Vessels (2 switched vessels)
3. **T**ricuspid atresia (Tri = 3)
4. **T**etralogy of Fallot (Tetra = 4)
5. **T**APVR (5 letters in the name)

Hypoplastic Left Heart Syndrome

Ebstein Anomaly

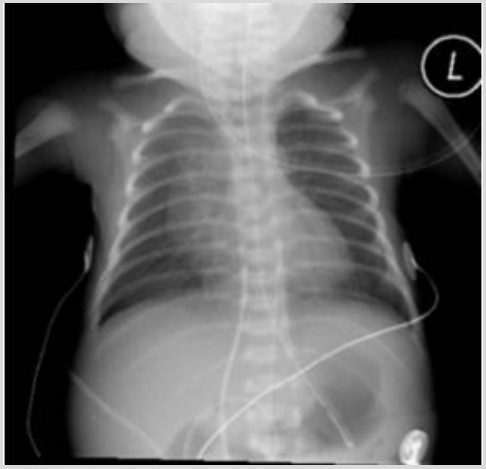
X-Rays in Cyanotic Heart Disease (CHD)

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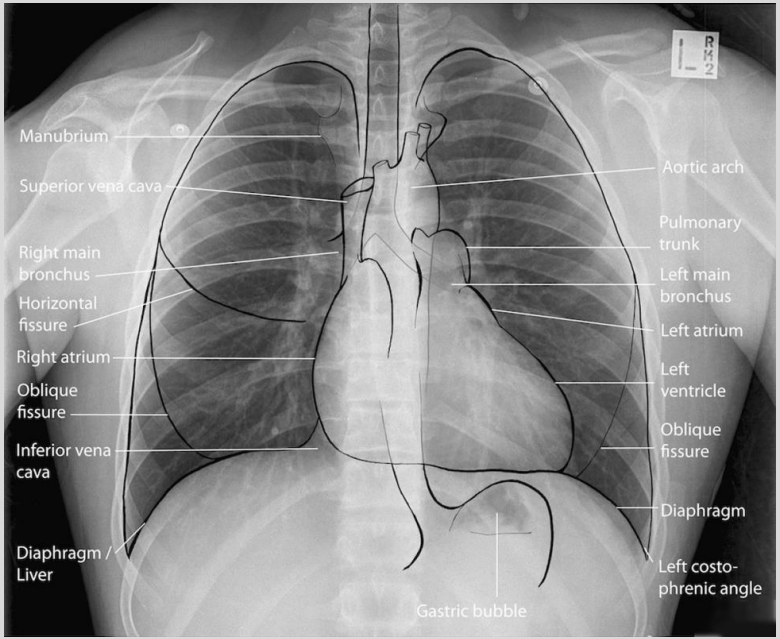
- Heart Size and Shape
- Increased Pulmonary Blood Flow (more White)
- Decreased Pulmonary Blood Flow (more Dark)



Increased pulmonary blood flow



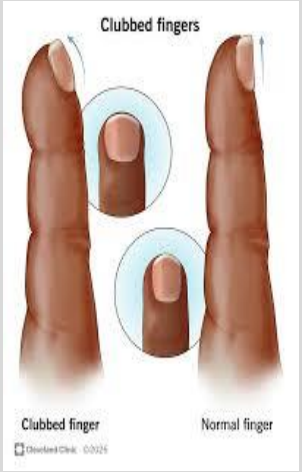
Decreased pulmonary blood flow



General Clinical Features of Cyanotic Congenital Heart Defects

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- Blue Babies
  - Or Pale, grey
- Feeding issues
  - Sweating while feeding
- Failure to thrive
- Tachypnea
- Fatigue
- Hypoxemia
- Nail clubbing later in life



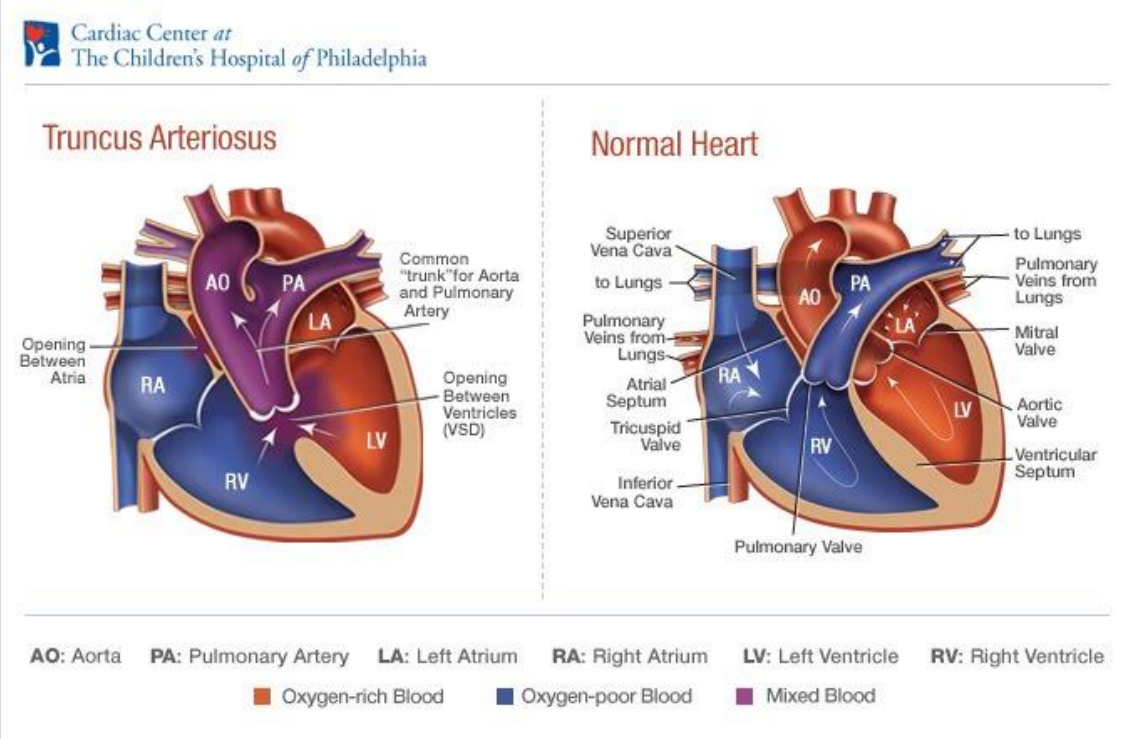


https://www.amboss.com/us/knowledge/Pulmonary_examination/

Persistent Truncus Arteriosus (1 vessel)

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Pathophysiology: absent or incomplete partitioning of the truncus arteriosus by the aorticopulmonary septum



https://www.chop.edu/img/cardiac-center/truncus-arteriosus.html

Persistent Truncus Arteriosus

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Mild cardiomegaly
Increased pulmonary markings

Treatment

- Surgical correction is recommended in the neonatal period

Associations:

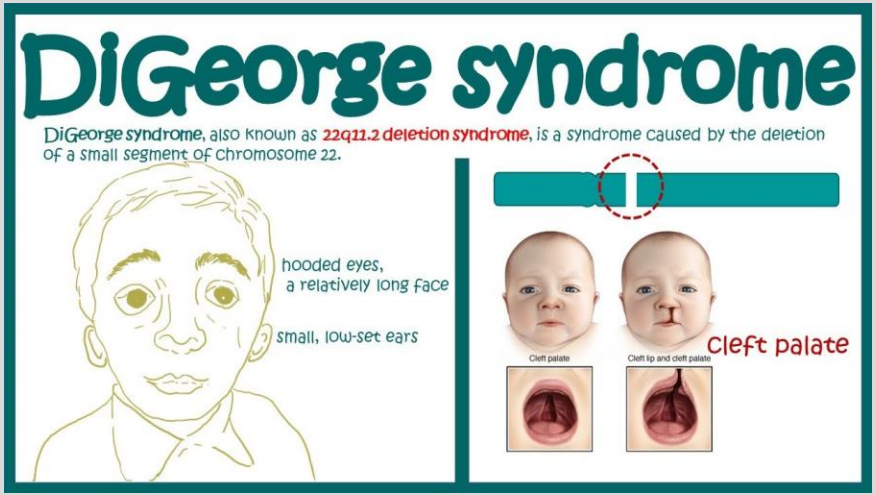
- DiGeorge Syndrome

Persistent Truncus Arteriosus

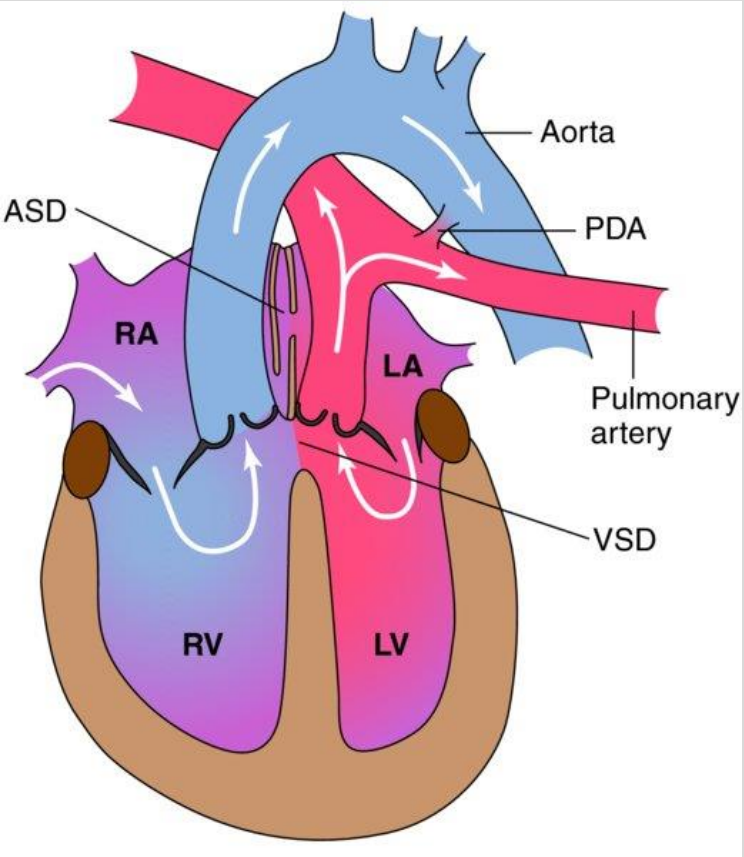
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DiGeorge Syndrome

- Conotruncal anomaly
  - Tetralogy of Fallot
  - Persistent Truncus Arteriosus
- Hypocalcemia (Sz, tremors)
- Absent thymus (immune defect)
- Facies/ low set ears
- Cleft palate
- Ch 22q11.2 deletion



Transposition of the Great Vessels

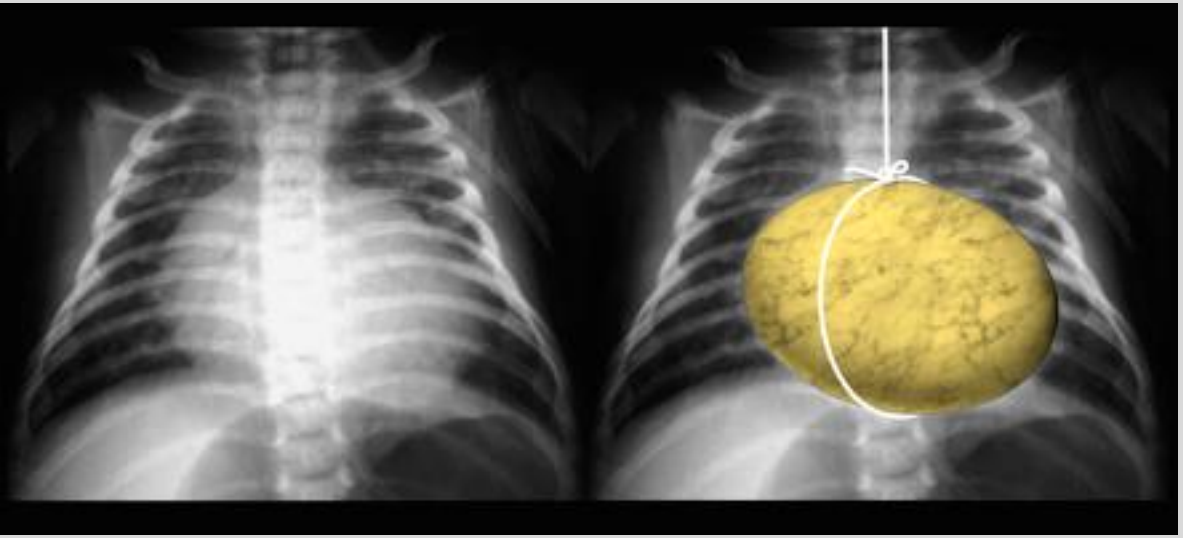


Pathophysiology: Failed spiraling of aorticopulmonary septum

https://meded.lwwhealthlibrary.com/content.aspx?bookId=2766&sectionId=231397027&resultClick=1#231397107

Transposition of the Great Vessels

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"Egg on a string" CXR
Narrow mediastinum, globular heart

Treatment

- Prostaglandin infusion
- Rashkind balloon – atrial septostomy
- Jatene procedure - Arterial switch operation within the first two weeks of life

Associations:

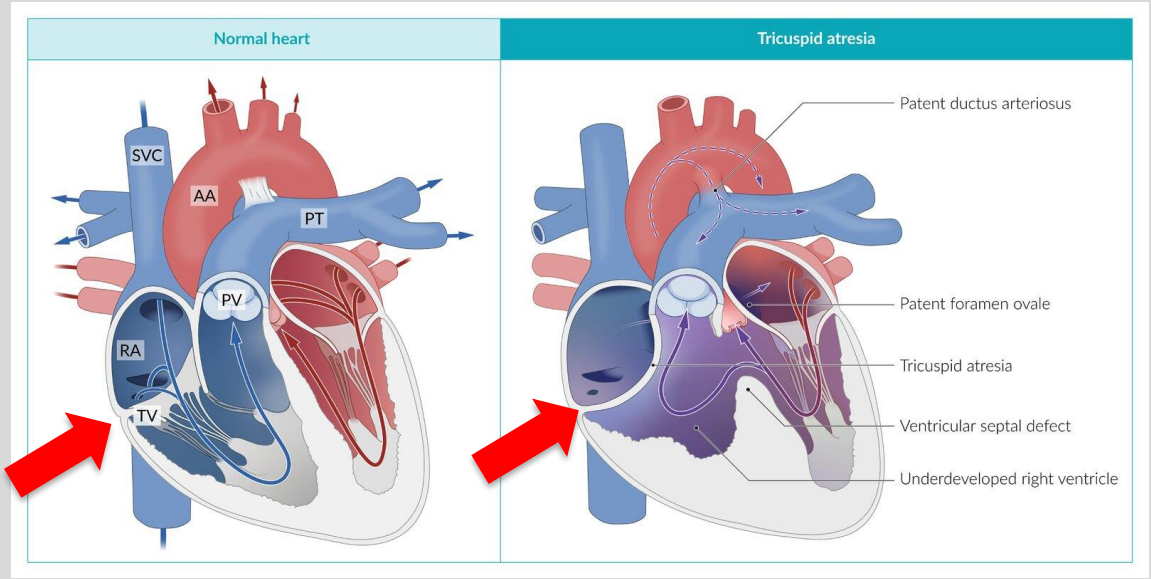
- **D**iabetic mom

https://radiopaedia.org/articles/egg-on-a-string-sign-heart

Tricuspid Valve Atresia

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- **Pathophysiology**: Absence of tricuspid valve and hypoplastic right ventricle
- Right atrial dilation
- Requires both ASD and VSD for viability



Treatment:

- Surgery
- Prostaglandin to maintain PDA

https://www.amboss.com/us/knowledge/Cyanotic_congenital_heart_defects/

Tetralogy of Fallot

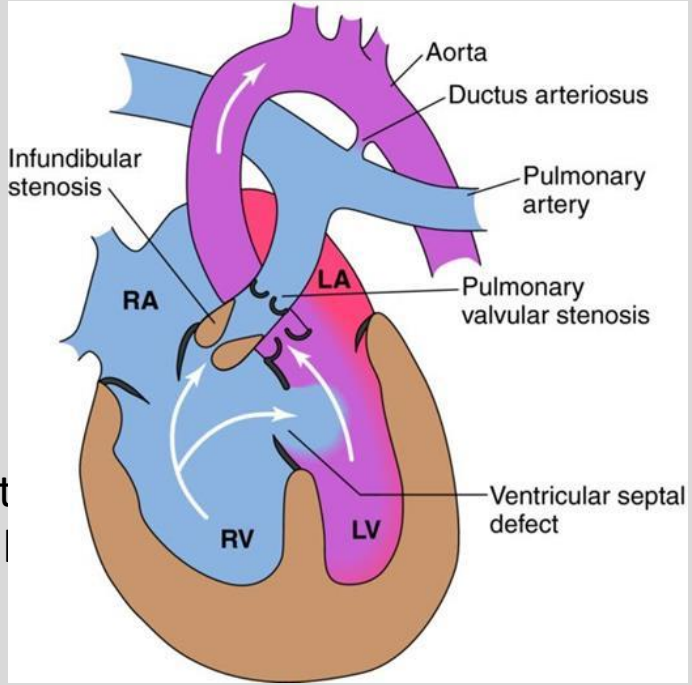
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- **Pathophysiology**: Caused by anterosuperior displacement of the infundibular septum

  - 1. Pulmonary artery stenosis
  - 2. Right ventricular hypertrophy (RVH)
  - 3. Overriding aorta
  - 4. Ventricular septal defect

  - MCC of early childhood cyanosis
  - Severity of the Pul stenosis determines the extent shunting. Other factors systemic resistance vs Pul vasc resistance

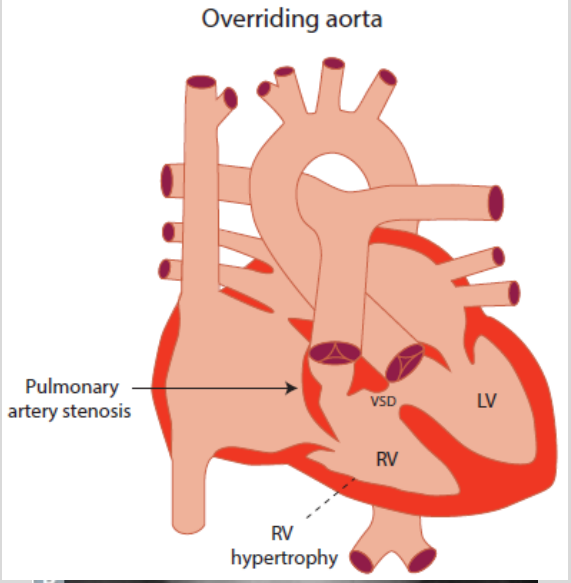
    "PROVe them wrong"



https://meded.lwwhealthlibrary.com/content.aspx?bookId=2766&sectionId=231397027&resultClick=1#231397107

Tetralogy of Fallot (cont.)

- Symptoms: failure to thrive, variable cyanosis, clubbing
- Tet Spells: Exacerbation of RV outflow obstruction or increase of pul pressures (exercise, crying, stress, etc) → ↑ R➔L shunting → ↑cyanosis
  - **Squatting** → ↑ SVR → ↓ R➔L shunt → ↑ pulmonary blood flow, ↑ oxygenation





First Aid for the Pediatrics Clerkship
First Aid Organ Systems

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Squatting Position



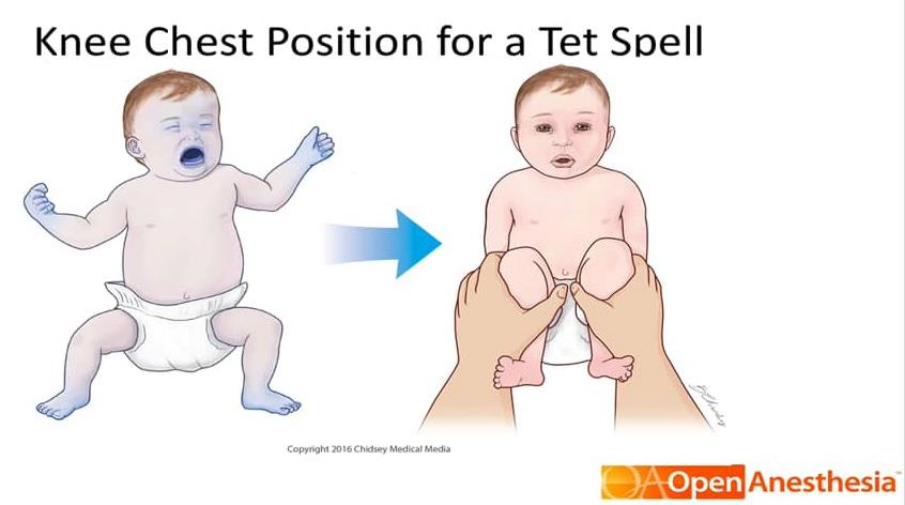


Tetralogy of Fallot

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Clinical Findings

- Tet spells: intermittent hypercyanotic, hypoxic episodes
- Associated with psychological and physical stress (crying, feeding, defecation)

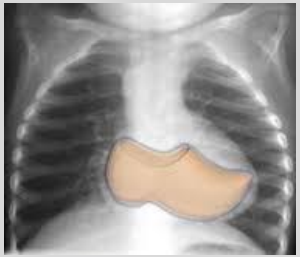


↑ PVR or ↓ SVR → ↑ Right-to-left shunting = "blue baby"

SQUAT = ↑ SVR → ↓ Right-to-left shunting

Tetralogy of Fallot





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Chest X-ray = Boot shaped heart
-Upturned Cardiac apex

Normal or decreased pulmonary vascular markings

Treatment

- Knees to chest position
- Oxygen, occ morphine
- Meds to increase systemic resistance (phenylephrine)
- Prostaglandin E (if PDA dependent)

Associations:

- EtOH exposure in utero
- Maternal rubella
- 22q11 syndromes
- Down syndrome

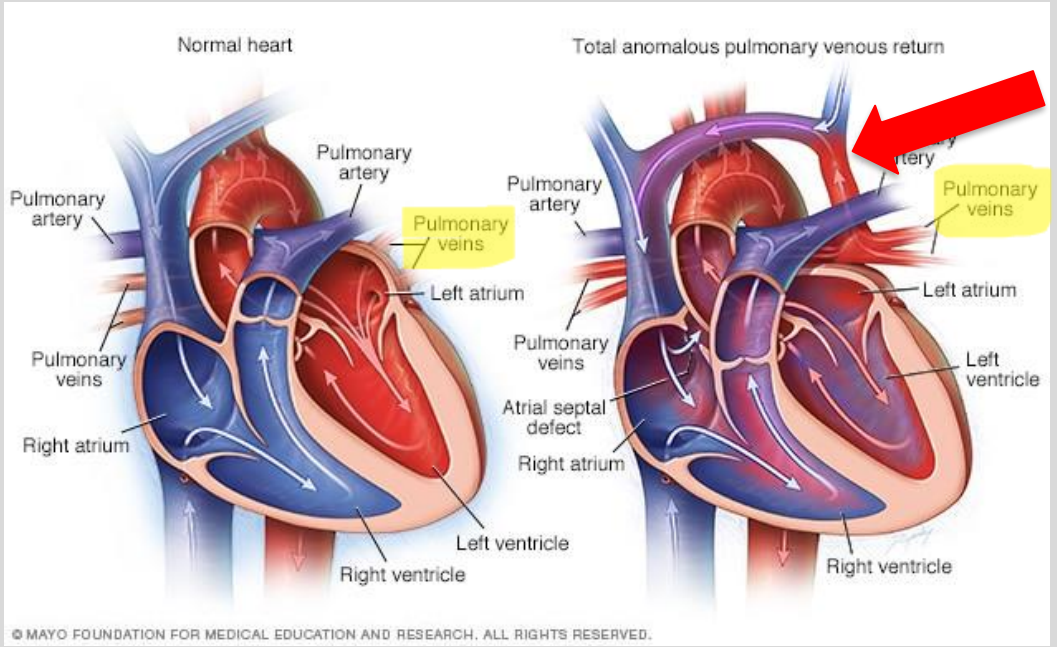
https://radiopaedia.org/cases/tetralogy-of-fallot-6

Total Anomalous pulmonary venous return (TAPVR)

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Pathophysiology: pulmonary veins drain into the right heart circulation

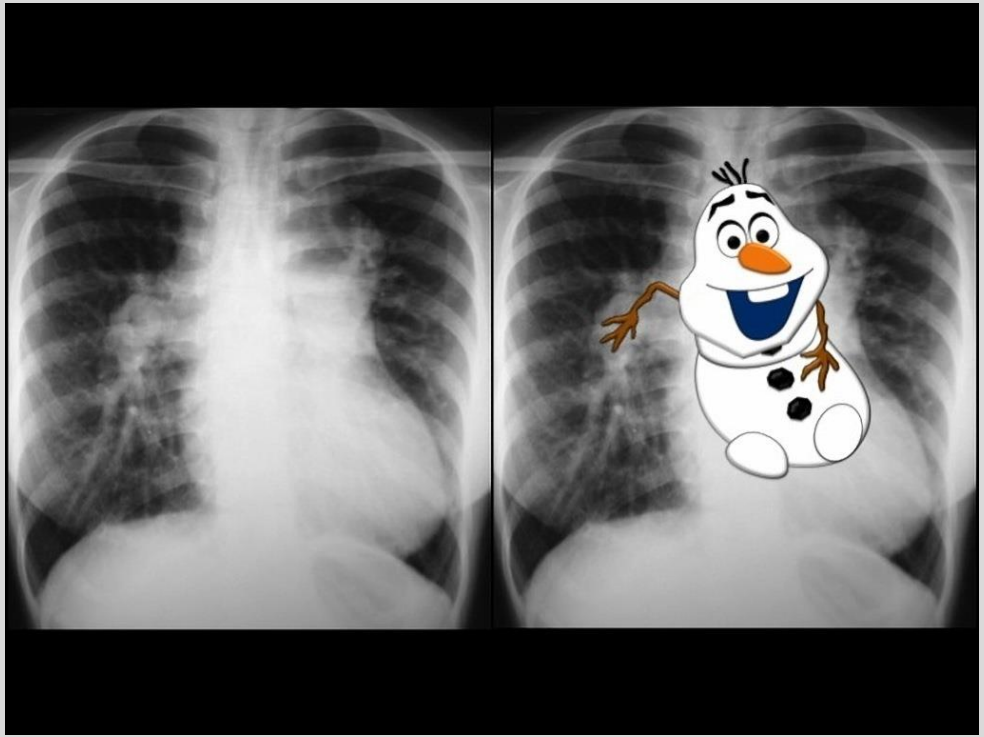
Need ASD +/or PDA



https://www.mayoclinic.org/diseases-conditions/total-anomalous-pulmonary-venous-return/cdc-20385613

Total Anomalous pulmonary venous return (TAPVR)

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Snowman in a snowstorm

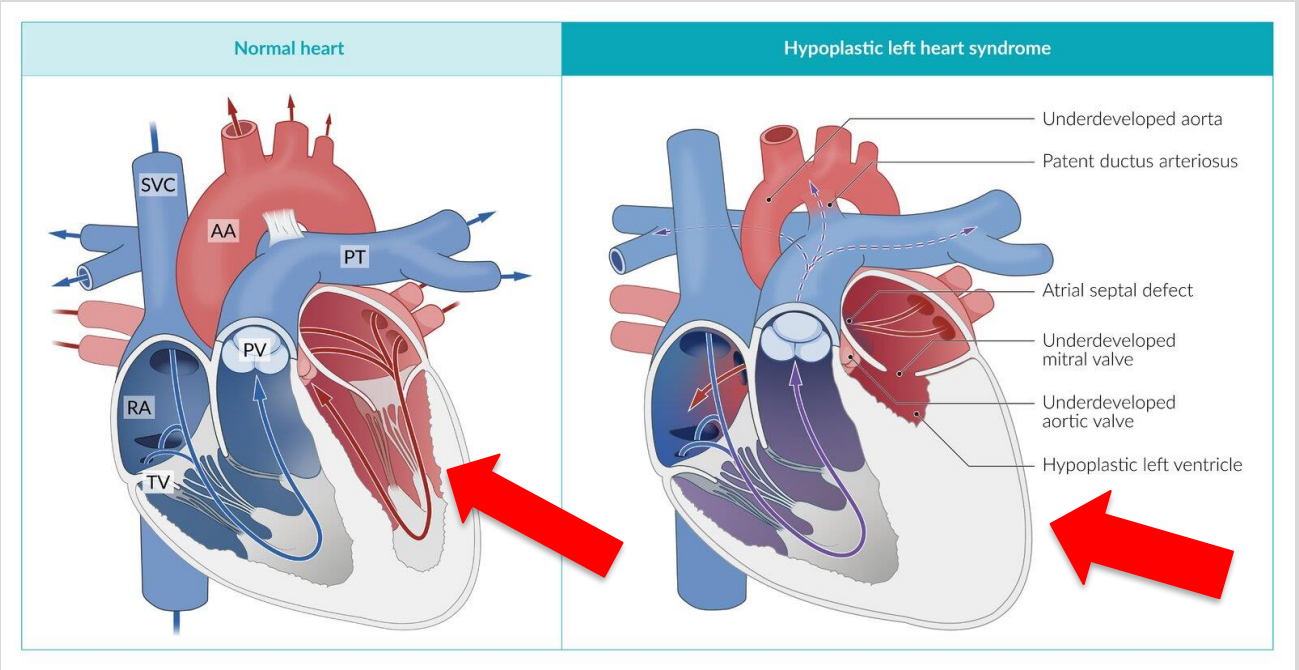
-Significantly increased vascular markings

Treatment:

- Surgery
- ASD necessary for survival
  - PDA allows for oxygenated blood to enter systemic circulation
  - Prostaglandin to help PDA stay open

Hypoplastic left heart syndrome

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Pathophysiology:
left side of the heart is hypoplastic/nonfunctional

Clinically not always cyanotic, grey, lethargic, looks septic, low O2 sat

Treatment: Prostaglandins, surgery

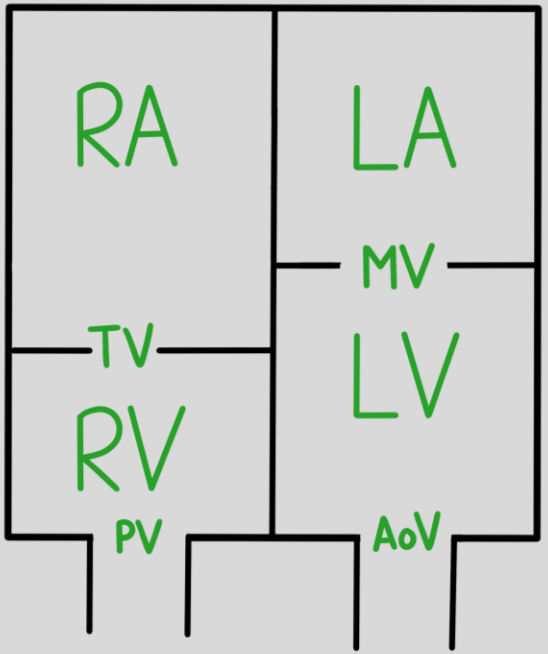
Need to PDA to get blood to systemic circulation

https://www.amboss.com/us/knowledge/Cyanotic_congenital_heart_defects/

Ebstein Anomaly

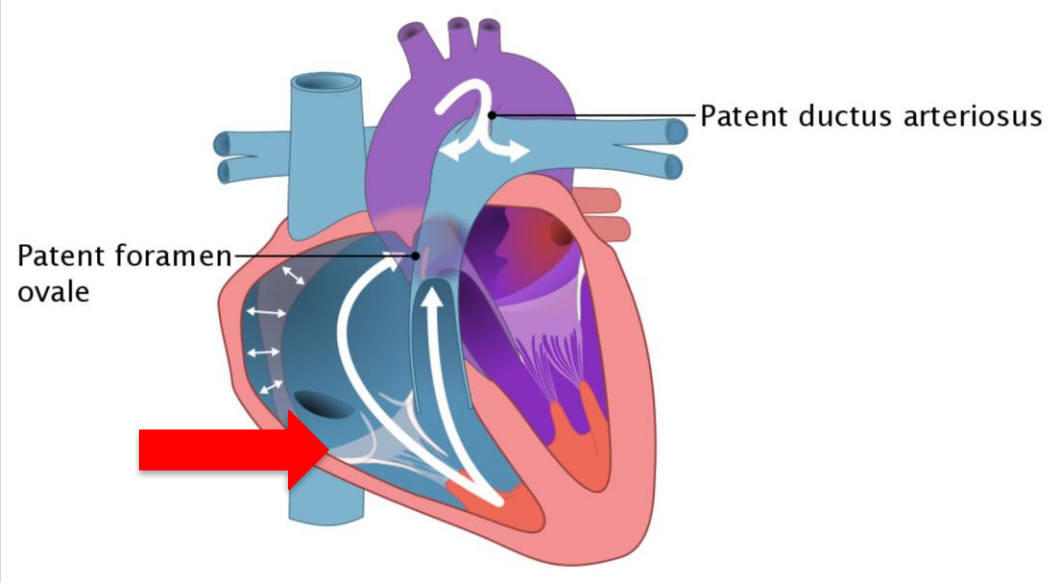
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- Tricuspid leaflets displaced into RV → "atrialization" of RV
  - A/w tricuspid regurgitation and R-sided heart failure
  - ASD often present (R to Lt shunt when Rt pressure builds)
- Symptoms: heart failure, cyanosis, respiratory distress
- Dx: Significant cardiomegaly on CXR, ECHO
- A/w **Lithium exposure** in utero
- Treatment: PGE to maintain PDA.



Ebstein Anomaly

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Pathophysiology: malformed tricuspid leaflets, and lower position
Small RV, larger RA

Associations: prenatal lithium exposure

ASD often present leading to R to L shunt
Prostaglandin to help PDA stay open

https://www.amboss.com/us/knowledge/Cyanotic_congenital_heart_defects/

Ebstein Anomaly

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Significant Cardiomegaly,
often decreased vascular markings
from poor blood flow to the lungs

Question #1

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A newborn girl is born to a 25-year-old mother with a past medical history of diabetes. On the second day of life, the infant is noted to be mildly cyanotic with an oxygen saturation of 80%. She is progressively more lethargic than usual with poor feeding. On physical exam, she has cyanosis, tachypnea, sternal retractions, and nasal flaring. A chest radiograph shows a narrowed mediastinum and a globular cardiac shadow. What is the most likely diagnosis?

a. Tetralogy of Fallot
b. Transposition of the Great Vessels
c. Tricuspid Atresia
d. Persistent Truncus Arteriosus

Question #1

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A 1-year-old presents to the emergency department with shortness of breath. His mom states that he turned blue after having a temper tantrum. The infant is noted to be mildly cyanotic with an oxygen saturation of 86%. His heart rate is 200 (normal =80-130), respiratory rate 36 (normal = 25-35). Below is the child's chest x-ray:

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What is the most likely diagnosis?

a. Total Anomalous Venous Return
b. Tetralogy of Fallot
c. Persistent Truncus Arteriosus
d. Tricuspid Atresia

Question #2

A 1-year-old presents to the emergency department with shortness of breath. His mom states that he turned blue after having a temper tantrum. The infant is noted to be mildly cyanotic with an oxygen saturation of 86%. His heart rate is 200 (normal =80-130), respiratory rate 36 (normal = 25-35). Below is the child's chest x-ray:

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What is the most likely diagnosis?

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b. Tetralogy of Fallot
c. Persistent Truncus Arteriosus
d. Tricuspid Atresia

Question #2

https://radiopaedia.org/cases/tetralogy-of-fallot-6

Bonus Q

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A mother brings her 2-year-old male child with Down Syndrome in for evaluation. She notes that he sometimes turns blue-ish after running or crying, and seems to console himself by squatting down. Physical examination reveals loud systolic ejection murmur. Why does the child squat to alleviate his symptoms?

A) Decreases energy expenditure

B) Increases afterload to decrease right to left shunt

C) Decreases afterload to increase right to left shunt

D) Increases preload to increase right to left shunt

E) Decreases preload to increase left to right shunt

Bonus Q

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E) Decreases preload to increase left to right shunt

Important Associations

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Disorder	Defect
Alcohol Exposure in Utero	VSD, PDA, ASD, Tetralogy of Fallot (TOF)
Congenital Rubella	PDA, pulmonary artery stenosis, septal defects
Down Syndrome	AV septal defect, VSD, ASD, TOF
Diabetic mother	Transposition of the Great Vessels, VSD
Prenatal Lithium Exposure	Ebstein Anomaly
Turner Syndrome	Coarctation of the Aorta
22q11 Syndromes (eg DiGeorge)	Truncus arteriosus, Tetralogy of Fallot

	Truncus Arteriosus	Transposition of Vessels	Tricuspid Atresia	Tetralogy of Fallot	TAPVR	Hypoplastic Left Heart	Ebstein Anomaly
Pulmonary Blood Flow	Increased	Increased	Decreased	Decreased	Increased	Increased	---
Abnormal?	Common trunk, VSD	Two separate circuits, PDA/VSD	Tricuspid valve absent	VSD, overriding aorta, RVH, Pulmonary stenosis	Pul Veins feeding back into RA	Underdeveloped LV	Abnormal tricuspid valve formation-downward displaced
CXR	----	Egg on a string	Round	Boot Shaped	Snowman	----	---
EKG	Normal axis	RVH, RAD	LVH	RVH, RAD	RVH, RAH	RAH, RVH	RAD
Auscultation	Holosystolic murmur	Soft systolic ejection murmur MLSB	Harsh systolic murmur LSB	Harsh systolic ejection murmur	Soft systolic ejection murmur LSB	Soft mid systolic murmur LSB	Holosystolic murmur LSB
Other	22q11 syndromes	Maternal diabetes	---	Tet spells, 22q11 syndromes	---	---	Lithium exposure

References

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Dudek RW. BRS Embryology. 6th ed. Baltimore, MD: Wolters Kluwer; 2014 .

Le, Tao; Bhushan, Vikas; and Sochat, Matthew. **First Aid** for the **USMLE Step 1 2021**. New York: McGraw-Hill Education, **2021**

https://www.amboss.com/us/knowledge/Prenatal_and_postnatal_physiology/

https://www.chop.edu/img/cardiac-center/truncus-arteriosus.html

https://radiopaedia.org/cases/tetralogy-of-fallot-6

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Feedback form:
https://comresearchdata.nyit.edu/redcap/surveys/?s=HRCY448FWYXREL4R